PACLITAXEL

Hypersensitivity — premedicate
Peripheral neuropathy

CHEMICAL NAME Taxane diterpenoid from *Taxus brevifolia* ($C_{47}H_{51}NO_{14}$)
CLASS Taxane (microtubule-stabilising) cytotoxic
TRADE NAMES Taxol · Abraxane (nab-paclitaxel) · Paclitaxel injection

THERAPEUTIC USES / INDICATIONS

- Breast cancer (adjuvant, metastatic).
- Ovarian cancer (with carboplatin).
- Lung (NSCLC), gastric, oesophageal, cervical cancers; Kaposi's sarcoma.
- Nab-paclitaxel: pancreatic cancer.

ADMINISTRATION

- **IV** infusion over 1 h (weekly) or 3 h (3-weekly) via 0.22-micron in-line filter and non-PVC (PVC-free) set.
- Premedication 30–60 min before: dexamethasone, H1 antihistamine (chlorphenamine), H2 antagonist (famotidine).
- Nab-paclitaxel: no premedication needed; not interchangeable.

DOSAGES

- **3-weekly:** 175 mg/m² over 3 h.
- **Weekly schedules:** used in some protocols (check local regimen).
- **With carboplatin (ovarian):** 175 mg/m² + carboplatin AUC 5–6.
- Reduce for neuropathy, neutropenia, hepatic impairment.

PHARMACOKINETICS

- Highly protein-bound; metabolised by CYP2C8 and 3A4.
- Excreted in bile. **t½:** 3–52 h (dose-dependent).

PHARMACODYNAMICS

- Binds **β-tubulin** → stabilises microtubules (prevents disassembly) → mitotic arrest (M phase) → apoptosis.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Alopecia, fatigue.
- Myalgia, arthralgia (2–3 days after).
- Nausea, mucositis, nail changes.

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity reactions (Cremophor vehicle) — anaphylaxis, especially first 2 infusions.
- Peripheral sensory neuropathy (cumulative).
- Neutropenia (dose-limiting).
- Bradycardia, arrhythmias; pneumonitis.

CONTRAINDICATIONS & CAUTIONS

- Severe hypersensitivity to paclitaxel or Cremophor EL.
- Baseline neutrophils <1.5 × 10^9/L.
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **CYP2C8/3A4 inhibitors (gemfibrozil, ketoconazole, clopidogrel):** ↑ toxicity.
- **Doxorubicin:** give paclitaxel after doxorubicin (sequence affects toxicity).
- **Cisplatin given first:** ↓ paclitaxel clearance.

NURSING MANAGEMENT

- Confirm premedication given and timed correctly.
- Vital signs at start, 15 min, then regularly; stay with patient during first 15 min; anaphylaxis kit ready.
- Use filter and PVC-free set.
- Assess neuropathy before each dose; FBC.
- Scalp cooling option.

PATIENT EDUCATION

- Take steroid tablets at the prescribed times before treatment (if oral premedication).
- Report flushing, chest tightness, breathlessness during infusion immediately.
- Report numbness/tingling in hands or feet.
- Muscle aches after treatment — paracetamol helps.

KEY POINTS

- Taxane; stabilises microtubules.
- Hypersensitivity, neuropathy, neutropenia.
- Premedicate (dex + H1 + H2); filter, PVC-free set.
- Breast and ovarian cancer.

SOURCES CHECKED **[894]** Paclitaxel (Pazenir) 5 mg/ml Powder for Dispersion for Infusion (UK SmPC) **[895]** Paclitaxel Protein Bound Particles Albumin Bound, IV (US label) **[896]** Paclitaxel protein-bound particles for injectable suspension (albumin-bound), IV (US label)
Full references p. 432–457